CM3070 Project
Lecture 2

BSc and Diploma in Computer Science and Related Subjects

Synopsis for Lecture

- Reporting Project Work & Progress
- Developing the Literature Review
- The “ABCs” of Literature Review
- Final Things for this Session

Reporting Project Work & Progress

Assessment of the Project module (with formative feedback and peer-assessment)

Description	Coursera Week	Elements	Weightage
Project Proposal (Video)	4	• Project Concept and Related Work *(by 10 Nov 2025)*	Accepted / Not Accepted
Draft Literature Review (Report)	6	• 4-6 examples of previous work or academic literature related to the project	Formative; Peer-assessment
Project Design (Report)	8	• Overview of the project • Name which template used for the project; Domain and users of the project; Design choices, technologies and methods • Overall structure of the project; Project plan of your work including testing and evaluation	Formative; Peer-assessment
Preliminary Project Report	11	• Introduction: this will explain the project concept and motivation for the project • Literatures review: a revised version from peer review • Project Design: revised version from peer review • Feature Prototype: implementation of at least one of the most important technical features used in the project *(by 18 Dec 2025)*	10%
Project Prototype	12	• Written description of the prototype you are developing	Formative; Peer-assessment
Project Testing	14	• 3 questions for the reviewers to answer about your project	Formative; Peer-assessment
Draft Project Report	16	• Introduction: this will explain the project concept and motivation for the project • Literatures review: a revised version from peer review • Project Design: revised version from peer review • Implementation: major algorithms/techniques used, explanation of most important parts of code and a visual representation of results • Evaluation: unit testing, user studies or testing on data and give the results • Conclusion: Short summary of the project and suggest further work	Formative
Project Testing 2	18	• Revised submission of Project Testing	Formative; Peer-assessment
Draft Project Report 2	18	• Revised submission of Draft Project Report	Formative
Unseen Project Examination (2-hour written)	22/23	• Understanding of Project and How it was carried out	20%
Final Project Report and Code + Project Presentation Video	24	• Project presentation video and the final project report and code *(by 8 Mar 2026)*	65%
Check in Quizzes / Progress Logs	1-20	• Online Quizzes	5%
Total Weightage			**100%**

Literature Review (Coursera Week 6 Submission)

Developing the Literature Review (1/4)

- Coursera Week 6 requires students to submit a Literature Review for peer assessment. Before this, students are not required to submit any written content (apart from Video in Week 4).
- This literature review can include academic papers and books. It can also include online articles and websites, but in that case, students should also explain why you consider them credible sources.
- **Students should evaluate the literature and explain how it contributes to your project, or how it demonstrates the gaps that your project intends to fill.**

Developing the Literature Review (2/4)

- Students should submit a draft literature review for your project. This should be a document covering at least 4 to 6 examples of previous work or academic literature.

- Examples of types of literature to include:
  - **Similar Projects.** Examples of projects that are similar to your own (these can be the same as those used in your proposal).
  - **Techniques and methods** that you plan to use. These could be software libraries, algorithms, or research methodologies.
  - Research **studies that show the effectiveness of the project** you intend to create (for example, if you are doing a project on educational technology, a psychological or educational study showing that the techniques you are using are effective for teaching).

Developing the Literature Review (3/4)

- This literature can include academic papers and books. It can also include online articles and websites, but in that case, you should also explain why you consider them credible sources.

- Review the work and explain how it motivates the project - not simply how good their research is.

- You should evaluate the literature and explain how it contributes to your project, or how it demonstrates the gaps that your project intends to fill.

- The literature review should also mention what ideas and analysis come out of that and how do they apply to the project you are doing.

Developing the Literature Review (4/4)

- The report should be in PDF format, up to 6 pages (minimum font size 11pt, minimum margin 2cm).
- There is a **word limit** in the summative assessments **of 2500 words** for your literature review, so it is a good idea to try to keep within this limit.
- Students may include additional pages of images and citations.
- Students should, if appropriate, use visual materials, for example, images and diagrams from the literature you are citing.
- Students might also want to include tables that compare different examples of prior work.

Marking Criteria for Literature Review

- Does it display knowledge of the area of study, previous work and academic literature? This is to assess how much you understand the domain area from the review conducted.
- Does it critically evaluate the previous work and/or academic literature? This is to assess whether you are making an argument based on the literature for the techniques or methods you are going to use for the design and development of the project?
- Does it use proper citations? This is to assess whether you are using proper citation and academic references.
- Does the report add value beyond summarising the different work coherently? This is to assess whether you have synthesise the knowledge acquired from understanding the articles.

The “ABCs” of Literature Review

Understanding Literature & Literature Review

Collection of scholarly writings of a topic area

- *What is a Literature*

Critical examination of the Literature

- *What is a Literature Review*

Writing the Literature Review is a Process

- *How do I get Started*

What comprises a Literature Review

A literature review must do these things:

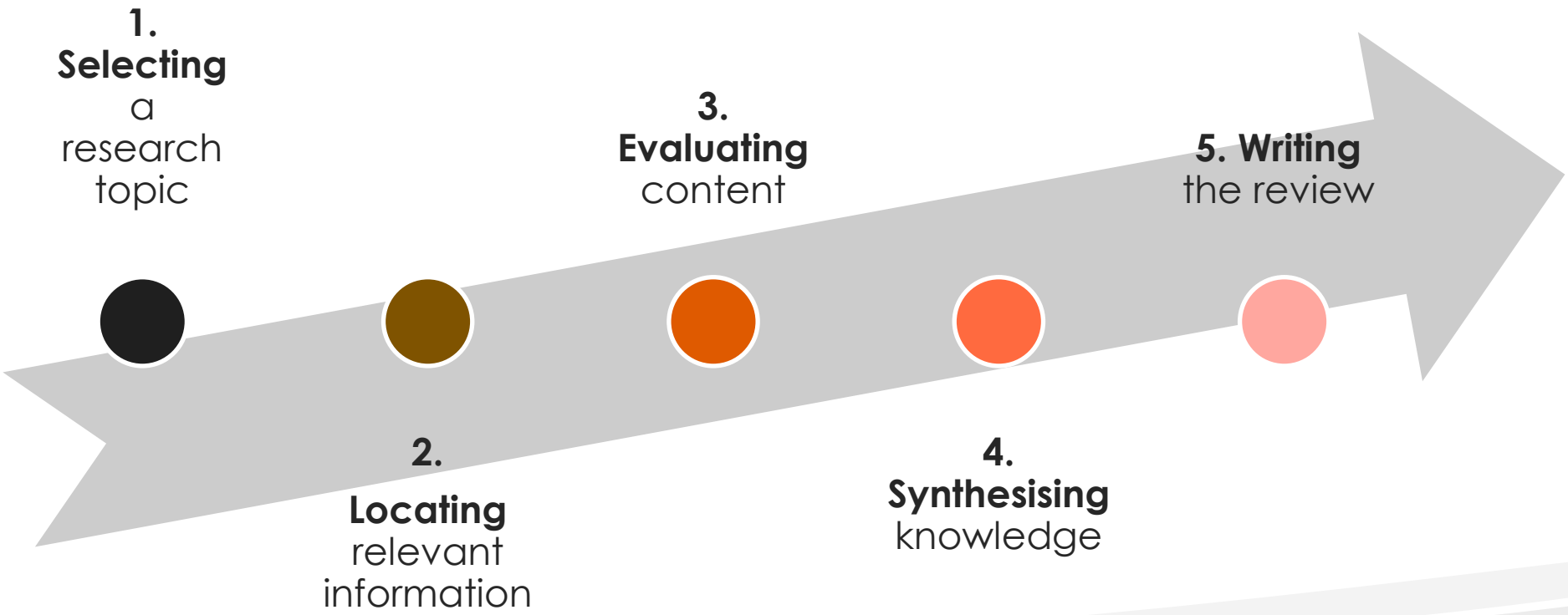
- Analyses information around and related directly to the topic area you are developing
- Synthesises results into a summary of what is and is not known
- Identifies areas of controversy in the literature
- Formulates questions that need further research

Why are Literature Reviews Important

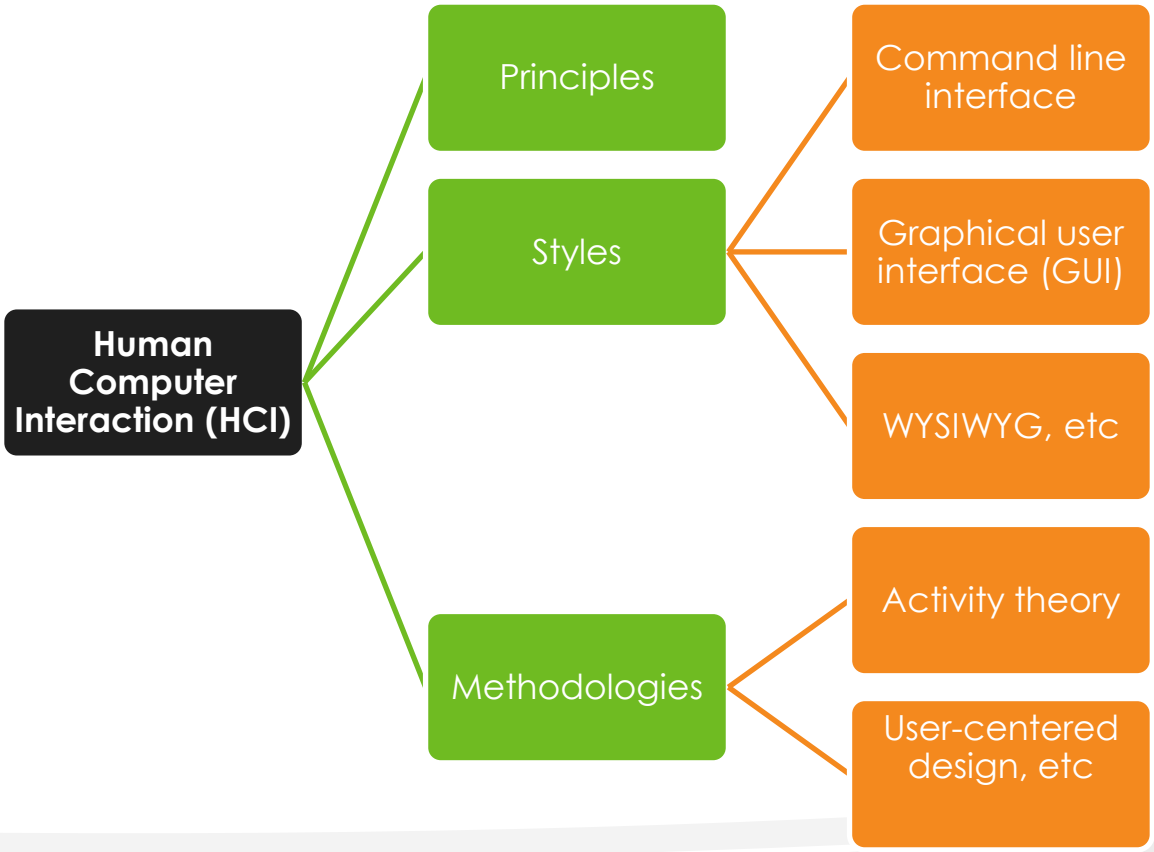
- A literature review:
  - Provides guide to the literature on a topic
  - Offers overview of previous work on a topic
  - Enlarges your scope of knowledge about a topic
  - Demonstrates command of the literature in your topic
  - Offers opportunity to conduct critical appraisal
  - Challenges your critical and analytical skills

- All of the above provides important information from body of knowledge you can use to conduct your project

Steps in Literature Review Process



1. Selecting a Research Topic



- Your research needs to be focused.
- For instance, although **HCI** sounds like a hot topic, it is too broad
- Think about how much literature has been published on such a general topic and how many themes we would encounter
- Instead, examine a narrow topic in greater details
- Zoom in on one aspect of HCI, eg. **the effect of different styles of HCI on the elderly**

Important Note: Your Literature Review

- Students should submit a draft literature review for your project. This should be a document covering at least 4 to 6 examples of previous work or academic literature.

2. Locating Relevant Information (1/4)

Books*	Journals*	Conference Proceedings
Government Publications*	Company Reports	Dissertations / Theses*
Manuals	Software	Internet / Social Media

*Must be main sources of information for Literature Review

2. Locating Relevant Information (2/4)

Creating a Search Strategy

Before you start, write a concise research question, which accurately reflects the focus of your study, eg. "HCI: An investigative study on the use of treemaps for blogshops"

Once you have determined your research question, identify the important terms, or **KEYWORDS**, which you will use to construct a search query. This is necessary because your research question, as stated, wouldn't make sense to an online search interface

2. Locating Relevant Information (3/4)

- Traditional Libraries – the public libraries, universities and SIM – *they have a large, systematically organised collection of books, media and journals*

- Databases – *We can search remote library catalogues or search several libraries simultaneously through electronic databases (must subscribe to them). One good source of search is* Online Library Athens from the UOL's Senate House Libraries: https://onlinelibrary.london.ac.uk/

  - ABI/Inform
  - Academic Search Premier
  - ACM Digital Library
  - Business Source Premier
  - LexisNexis Professional
  - Science Direct

2. Locating Relevant Information (4/4)

LASTLY, as you gather sources, it is important to keep in mind

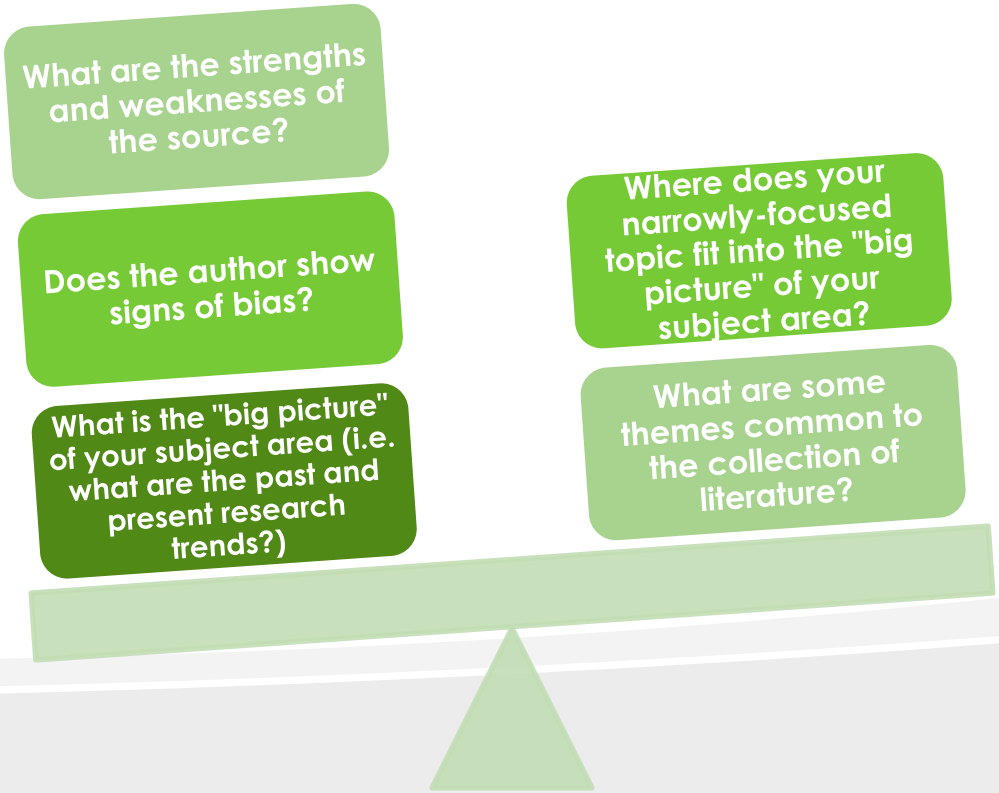
- Target the most significant literature in your field of study (you do NOT need to locate ALL the research published on your topic)
- Unless used for a historical perspective, sources should be current
- Focus on sources published by experts in the field
- Review references cited by experts as possible "leads"

Important Note: Your Literature Review

- Examples of types of literature to include:
  - **Similar Projects.** Examples of projects that are similar to your own (these can be the same as those used in your proposal).
  - **Techniques and methods** that you plan to use. These could be software libraries, algorithms, or research methodologies.
  - Research **studies that show the effectiveness of the project** you intend to create (for example, if you are doing a project on educational technology, a psychological or educational study showing that the techniques you are using are effective for teaching).

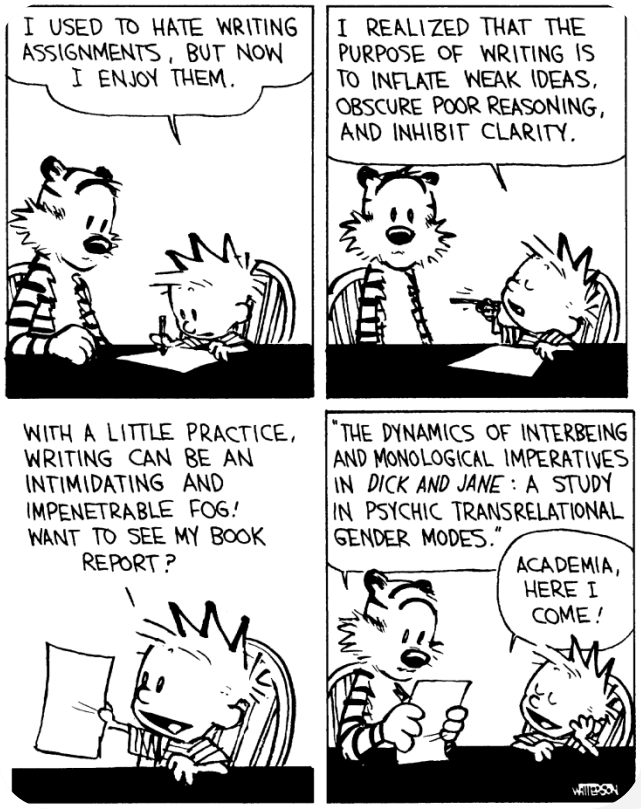
3. Evaluating Content (1/3)

As you go through this evaluation process, here are some important things to determine:

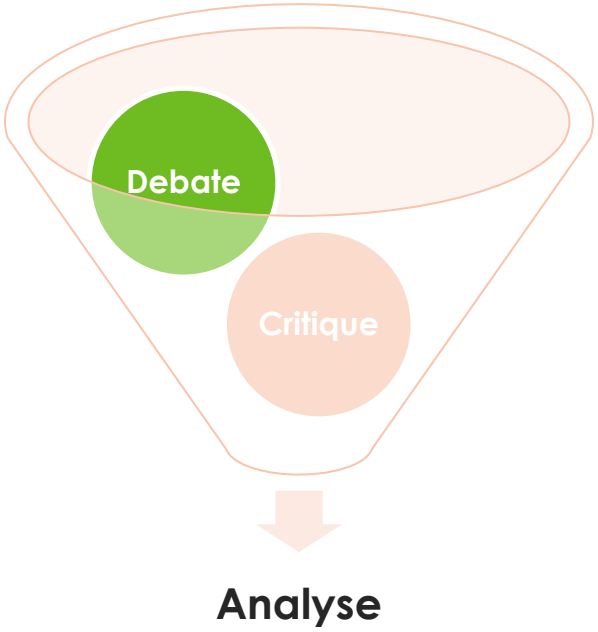


4. Synthesising Knowledge (1/4)

- Doing a search merely confirms whether the material you need is worth reading… that's all
- Still need to "digest" it - in order to formulate opinions and make decisions on how to use the material in the project report
- May need to read complex issues many times over before understanding them
- Good to read several articles about the same topic to provide scope for analysis and critique



4. Synthesising Knowledge (2/4)

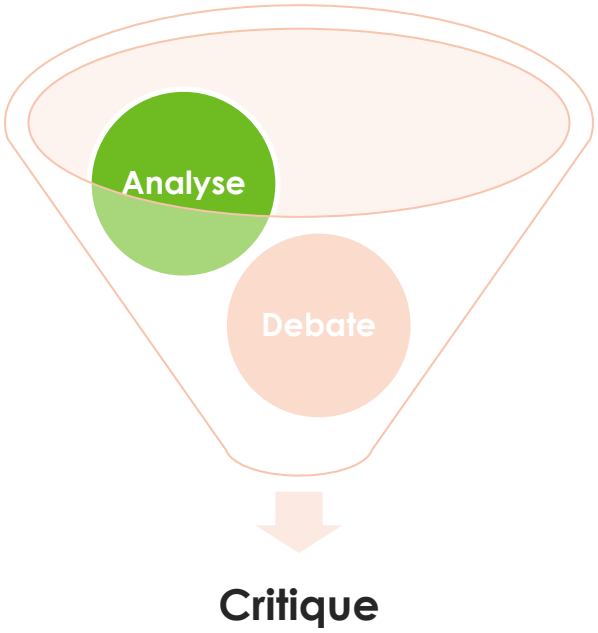


In your own words, analyse/summarise the key findings relevant to your study

1. What are the key arguments, characteristics, concepts, methodologies, and/or figures? What are the existing debates/theories?

2. Sample Language for "Analyse":
   - AA (2002) has demonstrated…was concerned with…
   - XX and YY (2010) compared algorithms for handling… Their work deals with…

4. Synthesising Knowledge (3/4)



Evaluate, compare, and critique the strength and weaknesses of the work

1. How do the different studies relate? What is new, different, or controversial? What evidence is lacking, inconclusive, contradicting, or too limited?

2. What research designs or methods seem valuable or unsatisfactory?

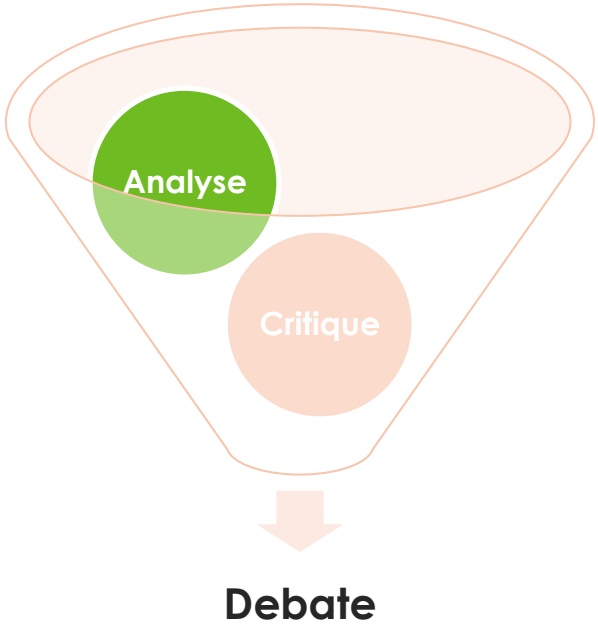
3. Sample Language for "Comparison/Critique":
   - In this ambitious but bias study, AA (2002)…
   - These results, reflecting the nature of the flow of goods, are similar to those reported by XX and YY(1999)…

4. Synthesising Knowledge (4/4)

Debate emphasises various strands of research in which critics/researchers explicitly engage with one another's work

1. Most useful when clearly differentiated positions are present in the literature

2. Sample Language for "Debate":
   - There have been <u>two (three, four, etc.) distinct approaches</u> this problem…The <u>first model</u> posits… The <u>second model</u> argues, conversely, that…



Important Note: Your Literature Review

- Review the work and explain how it motivates the project - not simply how good their research is. The literature review should also mention what ideas and analysis come out of that and how do they apply to the project you are doing.

5. Writing the Review (1/6)

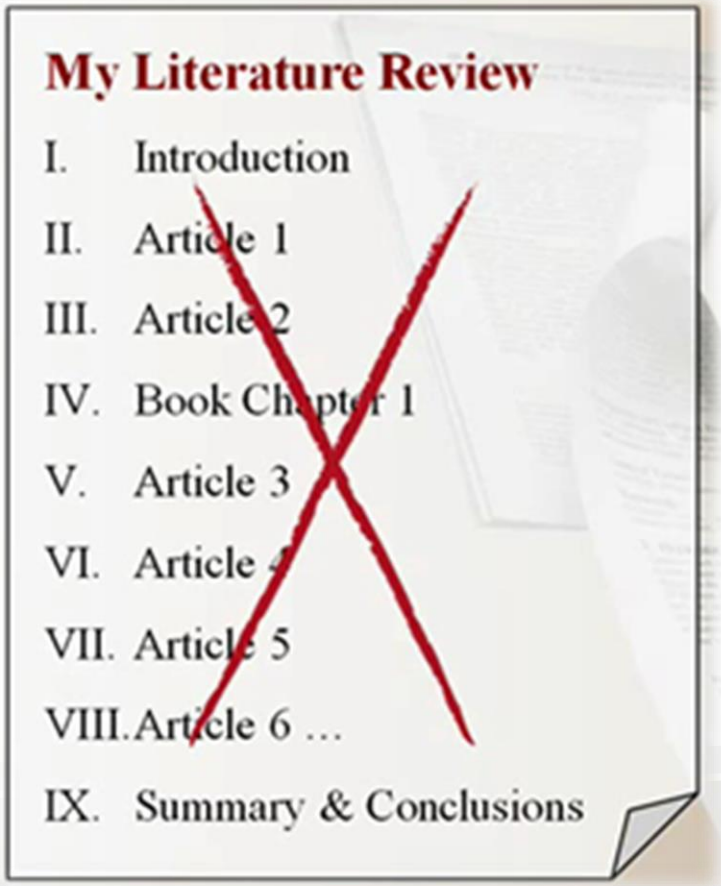
The final step is to write the Literature Review

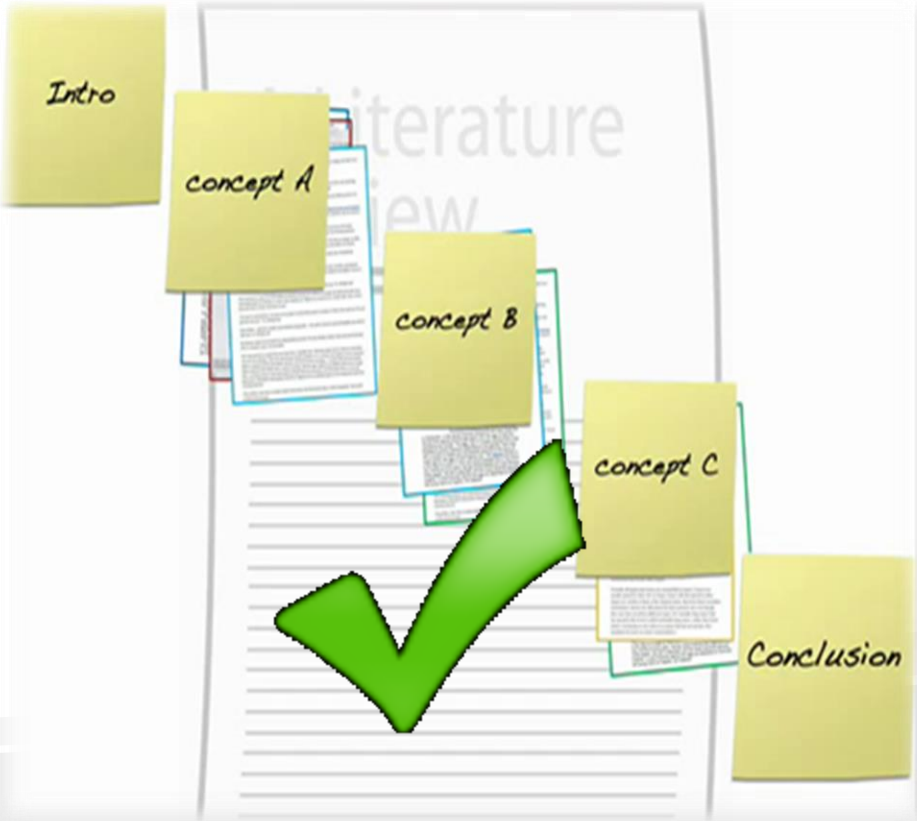
5. Writing the Review (2/6)

- Plagiarism – not referencing to sources
- Unable to identify or explain clearly the sources of information used
- Reference List not referenced or pointed to in the text of the Project Report
- Insufficient amount of literature survey and production of "deliverables by magic"

5. Writing the Review (3/6)

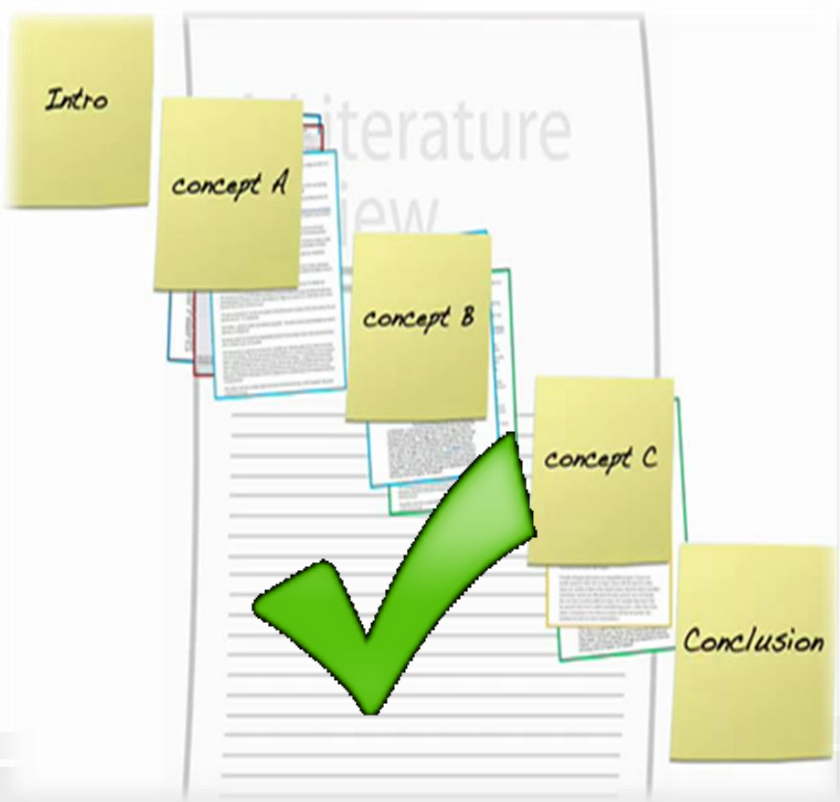
Structure of the Literature Review





5. Writing the Review (4/6)

Structure of the Literature Review



- Introduction begins by justification of content
- Shows how subject area has evolved over the years
- Chronological discussion of literature in the field
- Sets scene for reader – explains the area of study
- Subsequent focus on recent developments in research in the field
- Conclusion – limitations in the field mentioned and how the contents in the paper aims to fill the gaps

5. Writing the Review (5/6)

How long is the Literature Review

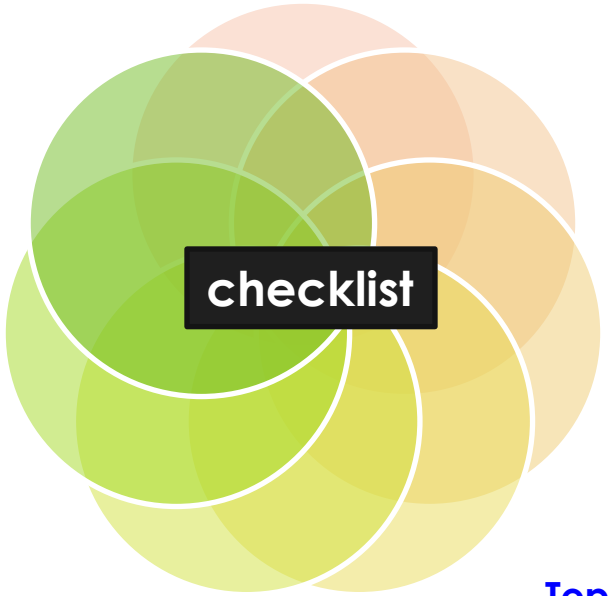
- The answer depends on the type of project
  - Research-Based, Evaluation and Problem-Solving Projects
  - Industry-Based and Development Projects
- Some students dedicate an entire chapter on Literature Review - Others may choose to embed the concepts/knowledge as the report is being written (like in most journal reporting) - this requires skills in doing it well
- Practice makes perfect - observe how it is done in printed materials and copy the technique, not the contents!

5. Writing the Review (6/6)

Topic: Is my title consistent with the content of my preview?

Spelling and Grammar: Are there any major spelling or grammatical mistakes?

Introduction: Do I appropriately introduce my review?



Conclusion: Do I provide sufficient closure?

Body: Is the organisation clear? Have I provided headings?

Sequence: Does my writing flow?

Topic sentences: Have I clearly expressed the major idea(s) of each paragraph?

Final Things for this Session

It’s not just about YOU, but US

DON’T BE AFRAID TO SEEK HELP

Acknowledgement

- Contents for this section were developed, adapted and/or referenced from the following sources:

1. http://writingcenter.unc.edu/handouts/literature-reviews/
2. https://student.unsw.edu.au/getting-started-your-literature-review
3. http://www.lib.ncsu.edu/tutorials/lit-review/
4. http://library.usm.maine.edu/tutorials/esp/how_to_tutorial.htm
5. http://www.wesleyan.edu/libr/guides/litrev/whatsalitreview.html
6. http://faculty.mwsu.edu/psychology/Laura.Spiller/Experimental/sample_apa_style_litreview.pdf
7. http://cisnet.baruch.cuny.edu/friedman/rm/277-8204.pdf

See you at next session !